

Dear Prof. Dr. Liesenborghs,

I am writing to express my strong interest in the PhD position on Immune Response and Antiviral Therapies for Mpox within the Virology, Antiviral & Vaccine Research group at the Rega Institute. This inspiring position perfectly matches my long-standing interest in translational biology and virology, and I am eager to apply my interdisciplinary training and hands-on laboratory expertise to advance our understanding of one of the most pressing viral threats of recent years.

I hold a BSc in Biology (with a GPA of 4.83 out of 5.00) and an MSc in Medical Biology (with a GPA of 7.63 out of 10.00), during which I received extensive training in molecular biology, genetics, physiology, and infectious diseases. My laboratory experience spans a broad set of techniques, including PCR/qPCR, western blotting, RNA/DNA extraction, ELISA, immunohistochemistry, and cell and tissue culture. During my time as a Junior Research Assistant, I studied the role of acid-sensitive two-pore potassium channels (TASK-1) in the vascular tone regulation in rat arteries *ex vivo* and the arterial regulation in the rat preeclampsia model using the wire myography, RNA extraction, qPCR, and western blotting. This experience taught me to work precisely with sensitive biological systems and to integrate different methodological approaches. I have also learned how to collaborate across interdisciplinary teams and to share results clearly through publications and presentations, which will help me communicate research findings effectively.

My motivation to join this PhD project stems from a genuine interest in virology and the societal impact of emerging infectious diseases. The COVID-19 pandemic demonstrated how critical it is to rapidly characterize pathogens, understand immune responses, and develop effective countermeasures. While assisting in the development and organization of (pre-)clinical trials at a Contract Research Organization, I saw the importance of translating molecular insights into tangible therapeutic advances. This experience reinforced my ambition to contribute to antiviral development. I am particularly drawn to this project because it not only investigates antiviral compounds but also addresses the immune mechanisms underlying mpox infection, which is key to developing lasting and broadly applicable therapies.

I am eager to contribute to the MBOTE project, collaborating with your team and international partners, while developing my own scientific creativity within this exciting research line. I thrive in collaborative, interdisciplinary environments and strongly value knowledge sharing. I am highly motivated to dedicate myself to a four-year PhD project that has both scientific and societal impact.

Thank you for considering my application. I look forward to the possibility of contributing to this exciting project.

Warm regards,
Varvara Lazarenko